

ROUTEX

AI-Powered Supply Chain Resilience

Turning disruption into a decision.

Detect

Understand

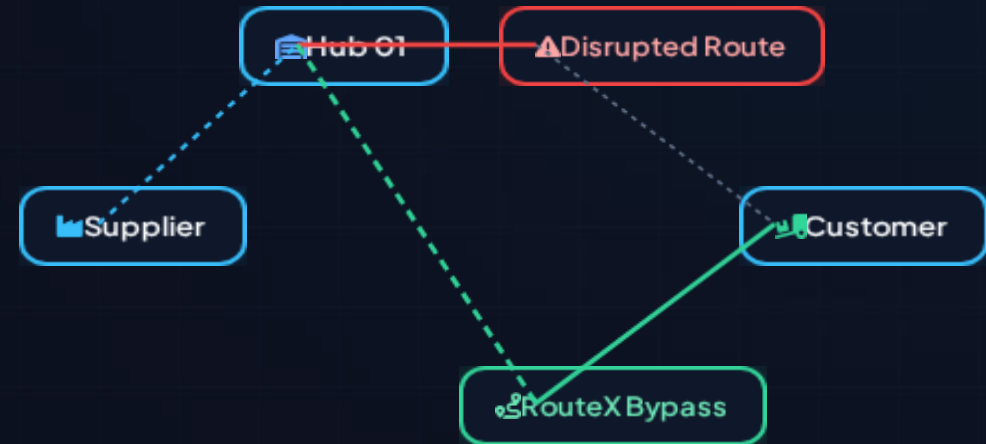
Optimize

Simulate

Act

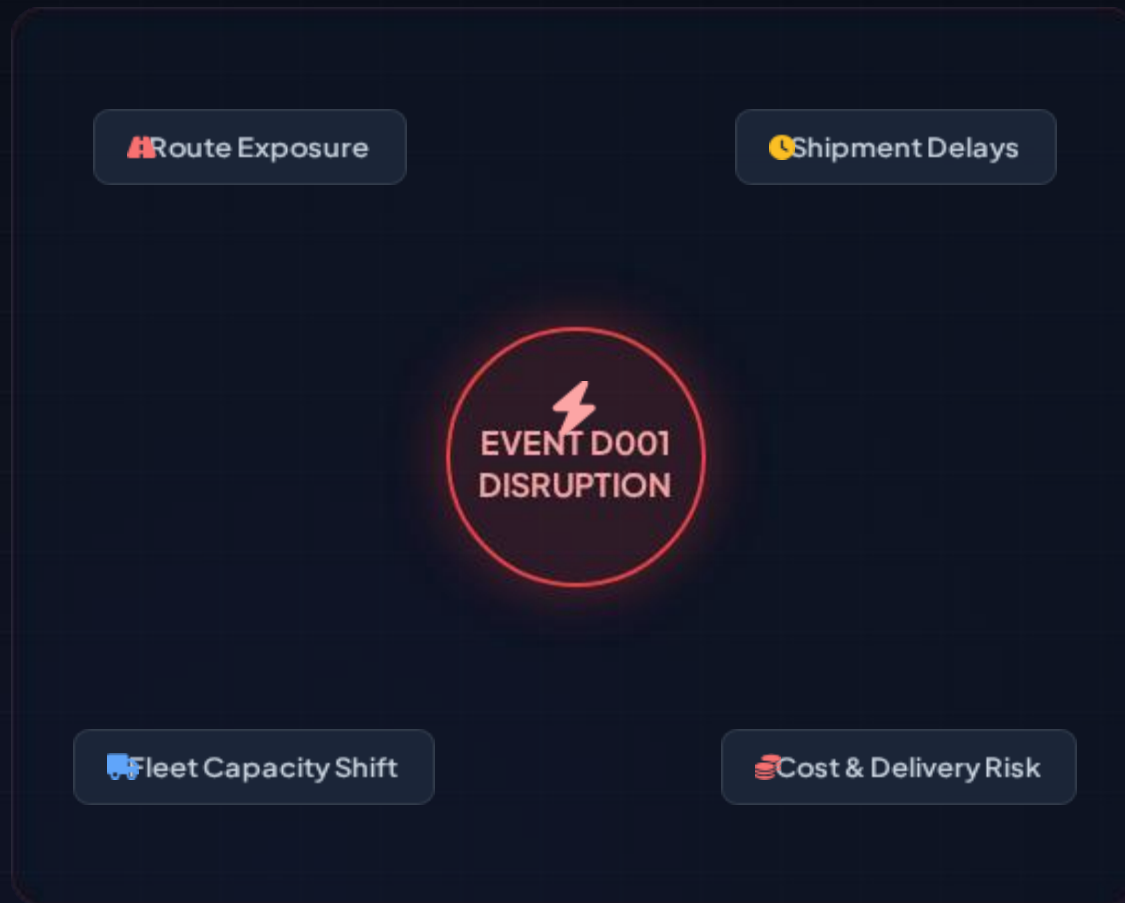


IBM Bob AI Hackathon Team
Presentation & Operational Copilot



Supply Chains Don't Fail Quietly

01 / PROBLEM



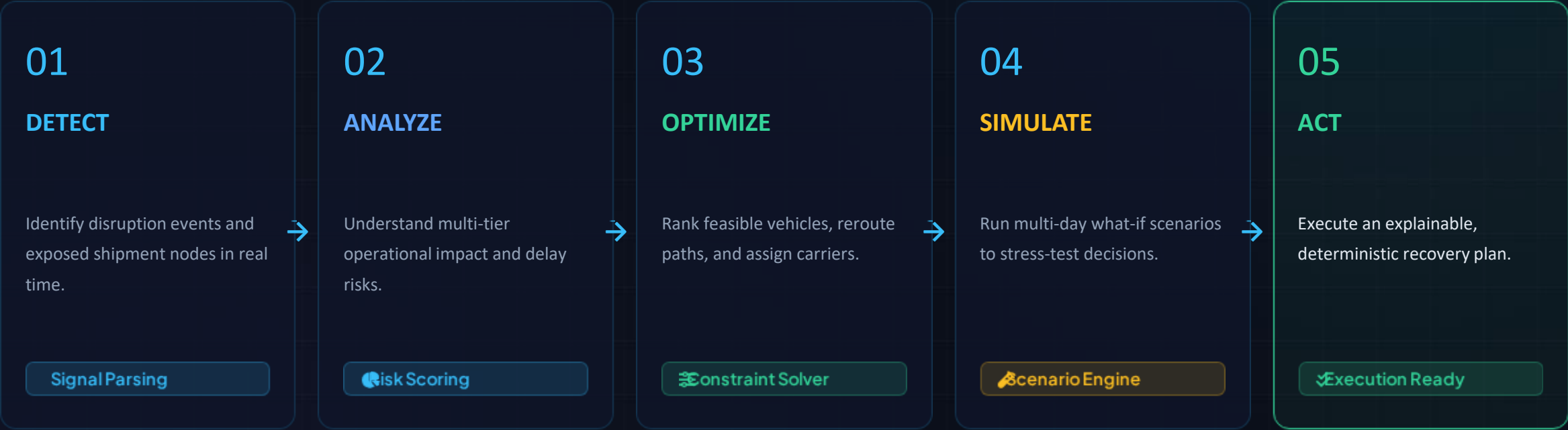
The Real Challenge

The real problem isn't detecting a disruption — it's knowing exactly what to do next.

- ✗ Routes become unavailable or highly risky instantly
- ✗ Shipments become exposed to escalating delay penalties
- ✗ Fleet capacity changes while carrier options become constrained
- ✗ Manual spreadsheets create a severe gap between alert and action

From Alert → Decision → Action

RouteX connects the entire disruption-response journey inside a single decision copilot.



From Event to Operational Picture

03 / INTELLIGENCE

Active Event Code: **D001 — Route Disruption Detected**

 **High Exposure Alert**



SHIPMENTS

Which cargo orders are directly affected in transit?



ROUTES

Which corridors carry unacceptable risk exposure?



FLEET

Which nearby vehicles are operational & available?



CAPACITY

Can available trucks physically accept the payload volume?



CARRIERS

Which partner carrier agreements remain compliant?



RISK SCORE

What is the quantified financial and SLA delay impact?

Instead of simply stating "*Route Disrupted*", RouteX answers: "**What is exposed — and what can we do?**"

Fleet Optimization Ranking

04 / FLEET ENGINE

Right Vehicle. Right Shipment.

RouteX does not simply pick the nearest available truck. It filters for strict operational feasibility and then ranks candidates by match score.



Location Proximity to origin

Capacity Cargo volume fit

Availability Duty status check

Compatibility Carrier compliance

 -014 Selected as Best Feasible Match



Choose the Best Recovery Path

05 / ROUTING

GPS CORRIDOR EVALUATOR

⚠️Primary Route A Blocked

Route A (Original)

520 km • 9 hrs • ₹18,000

HIGH EXPOSURE

✔️Route B (Recommended)

570 km • 10 hrs • ₹20,000

LOW EXPOSURE

Route C (Alternative)

610 km • 12 hrs • ₹17,000

MEDIUM EXPOSURE

👍Why Route B?

While Route B is 50 km longer than Route A, it avoids the active severe flood corridor, resulting in an **82% reduction in exposure risk** for critical payload.

RouteX evaluates supplied feasible alternatives balancing distance, duration, cost, and disruption exposure score.

Don't Wait for the Future. Simulate It.

SIMULATION SCENARIO: **Disruption D001** extended for 5 Days

BASELINE

Normal Ops

Exposed Cargo:**0 Units**

Fleet Status:**100% Standard**

Avg Delay:**0 Hours**

Cost Variance:**Baseline**



5-DAY DISRUPTION

Simulated Delta

Exposed Cargo:**+8 Shipments**

Fleet Required:**+2 Vehicles**

Avg Delay:**+14 Hours**

Risk Surcharge:**+12.4%**

Test decisions & reserve backup fleet before committing capital resources.

One Scenario. Multiple Possible Futures.

Deterministic scenario testing evaluates how operational response escalates across time horizons.



Move operators from *"Something might go wrong"* → **"Here is the playbook for each day."**

OPERATOR COMMAND

"What should we do if disruption D001 lasts for 5 days?"



ROUTEX RECOMMENDATION ENGINE

Explainable Logic

Reroute priority shipments via **Route B** and reallocate **2 feasible vehicles (V-014, V-022)** to maintain target SLA.

WHY? Feasible Capacity Verified + Proximity Advantage + 82% Exposure Reduction.



EXECUTE REROUTE REASSIGN FLEET

AI Recommends. Humans Remain in Control.

Modular Decision Engine

09 / ARCHITECTURE

 DATA & EVENTS

Shipments • Routes • Fleet Telemetry

 RISK ENGINE

Impact Assessment • Delay Exposure Scoring

 OPTIMIZATION

Fleet Matching • Route Selection • Carrier Assignment

 SIMULATOR

Baseline vs Scenario Delta Calculation

 AI COPILOT

Explainable Action Recommendations

 COMMAND CENTER

Operator Decision Execution



Modular Architecture Explainable Results Deterministic Rules REST API-Ready



Built for Operational Reliability

10 / TECH STACK

Backend & Optimization Services



Python-based optimization services & REST APIs



Deterministic business logic & constraint solvers



Structured JSON schemas for API contracts

Key Design Principle

Optimization logic is decoupled from AI explanation layers. The AI explains decisions without acting as the mathematical calculation engine.

```
// Endpoint Contracts
GET  /api/fleet
GET  /api/fleet/utilization

POST /api/optimize-fleet
POST /api/what-if

// JSON Optimization Payload
{
  "disruption_id"    : "D001" ,
  "duration_days"    : 5 ,
  "constraints"      : [ "capacity" , "carrier" ]
}
```

A Disruption Becomes a Decision in Seconds



01 DISRUPTION

Event D001 detected on main transit corridor



02 IMPACT

8 active shipments identified as exposed



03 OPTIMIZE

2 suitable trucks (V-014, V-022) matched



04 SIMULATE

5-day scenario stress test evaluated



05 ACT

Explainable recovery plan generated & executed

KEY OPERATOR QUERY "What should we do if D001 continues for 5 days?"

⚡ Action Plan Ready in < 2 Seconds

Reactive → Resilient Operational Shift

✗WITHOUT ROUTEX

Alert Received → Manual Spreadsheets → Static Planning → Delayed Reactive Response

✓WITH ROUTEX

Impact Quantified → AI Analysis → Resource Optimized → Scenario Tested → Proactive Recovery



Faster Response

Reduce manual decision time from hours to seconds



Better Visibility

Full clarity on exposed cargo, fleet & corridors



Resource Allocation

Use strictly feasible trucks, not just nearest trucks



Explainable Decisions

Audit-ready operational explanations for operators

ROUTEX

Turning disruption into a decision.

When the supply chain changes,
RouteX helps teams decide what to do next.



IBM Bob AI Hackathon
Thank You for Your Time

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